

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

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Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity that descends a squared-drift objective removes most of the objective available to it and keeps the system coherent, while driving within-cluster coupling far below between-cluster coupling—away from the

structure it is asked to learn, and further from it than from almost every relabelling of the nodes; a matched random update of the same step size does neither. Descending that functional does not produce representational learning, and its optimum is opposed to the target by a conservation law rather than by accident. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes and the two structures it takes alongside them, whose class fields carry physical content of their own; one witness assembled from distinct toy components satisfies all eight at once. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the

13 local descriptions that activity carries must be compatible enough to glue
14 into a single one; and that glued state must support an accurate model of
15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the

36 ramp at least as much as the mechanism—and that holds for any analysis
 37 reading a critical exponent off a driven transition, not only for ours. The
 38 third is a negative result about plasticity: a coupling update descending
 39 a squared-drift objective removes most of the objective available to it and
 40 preserves coherence, while moving *away* from the environmental structure
 41 it is expected to learn—below almost every node relabelling, and beyond
 42 what a matched random update of the same step size does. Descending a
 43 squared-drift functional does not produce representational learning, and the
 44 direction of that failure follows from a conservation law rather than from the
 45 particular run. The fourth is a machine-checked composition (Section 5) in
 46 which finite information capacity, Landauer’s bound, a predictive dissipation
 47 bound, the noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point
 48 theorem yield one conditional theorem, with the eight hypotheses that com-
 49 position consumes isolated and named; one witness assembled from distinct
 50 toy components satisfies all eight at once, so their conjunction is not empty.

51 A prose table of assumptions would list the same eight items, so the fourth
 52 contribution needs a word about method. What the Lean development adds
 53 is that it makes the list non-negotiable. The characteristic failure in an
 54 argument of this shape is not a false step but a silent change of subject:
 55 a theorem about a stationary density read as a claim about a trajectory,
 56 an inequality about a register read as a claim about learning, agreement of
 57 phases read as compatibility of representations. Type-checking forbids each
 58 of those substitutions, and an assumption needed to make a step go through

59 appears as a hypothesis in the term rather than as a word in a sentence. The
60 last of them is not merely forbidden but refuted by a constructed object: the
61 development carries a cover at a single common phase whose local densities
62 disagree where its patches meet, satisfying every other requirement of the
63 gluing theorem and failing compatibility alone (Section 5.2). That object
64 is what a prose table of assumptions cannot carry. Formalization does not
65 make the assumptions true; it makes them impossible to lose.

66 The physical model and the formal composition belong in one article be-
67 cause they are two halves of one commitment. The coupling-gated prediction
68 is worth testing under this framework, rather than as one more synchroniza-
69 tion model, only because the coherent state it produces is claimed to do
70 further work: to support a globally consistent representation and a reflexive
71 fixed point that we identify with phenomenal unity and a minimal self.

72 The distinction that does that work is between phases agreeing and de-
73 scriptions fitting together, and it is concrete. Take two cortical regions whose
74 receptive territories overlap, each carrying a probability distribution over
75 what occupies the shared part. Their oscillators can be locked to the same
76 phase at every instant while one region's distribution on the overlap places an
77 object left of fixation and the other places it right. The phases agree; the de-
78 scriptions do not, and no single distribution over the shared territory restricts
79 to both. A theory that identifies unity with synchrony cannot mark that dif-
80 ference, and so cannot separate a brain whose regions oscillate together from
81 one whose regions are about the same world. The sheaf condition marks

82 exactly it: local states glue to one global section when their restrictions to
 83 shared regions coincide, and otherwise not at all. Coherence enters as the
 84 physical condition under which compatibility can be maintained, not as a
 85 substitute for it, and the self-prediction map then asks the further ques-
 86 tion of whether the glued state contains an accurate model of itself. That
 87 identification is a commitment, not a theorem, and Section 6 states it, ar-
 88 gues for it, and says what would count against it. Its separation from the
 89 physical tests is what makes both assessable: the cortical coupling hypothe-
 90 sis and the recovery prediction can fail while the mathematical composition
 91 stands, and the composition can stand while the identification is wrong. The
 92 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;
 93 Banach, 1922) therefore appear here for their biological composition, not for
 94 any new fundamental physical law.

95 **2 A coupling-gated model of cortical coher-** 96 **ence**

97 **2.1 Dynamics and stationary solutions**

98 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

The I_n are modified Bessel functions, and ψ is the collective phase. Lean verifies normalization, the density's circular order parameter, and the self-consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent order parameter is zero. At $K > K_c$, exactly one positive solution coexists with zero; the positive branch is continuous at threshold and strictly increasing with K .

The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

This is a result about the stationary solution branch. Of the coherent branch we prove existence, uniqueness, continuous emergence at threshold and strict growth in K , but not that it is dynamically selected. The dynamical mean-field limit—propagation of chaos—is the missing link between the finite sys-

tem and the stationary description (Sznitman, 1991), and no theorem in this development closes it. Exchange symmetry alone does not justify factorization of the joint law, particularly for an ensemble with an unpinned collective phase.

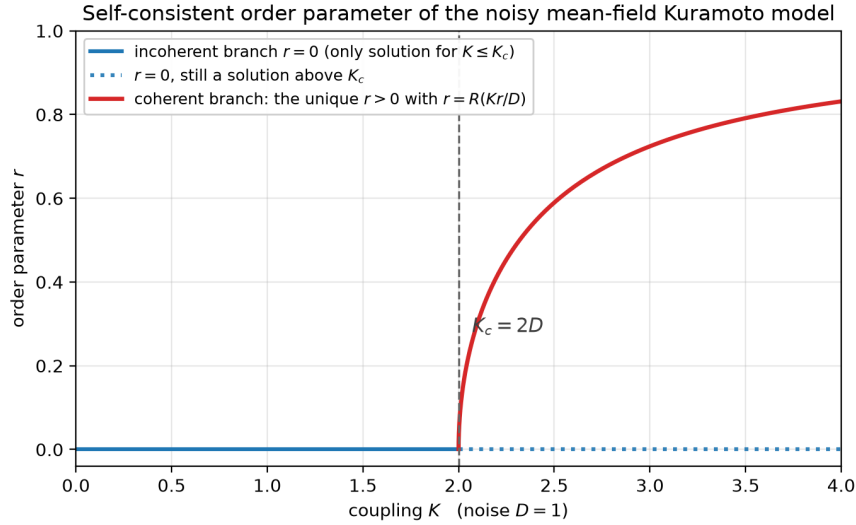


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

A spatial version replaces the finite sum by an integral against a kernel $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-field strength is a kernel average, not a quantity that grows with the arbitrary measure of the substrate. Reduction of a heterogeneous spatial kernel to a scalar K is an additional modelling step, and the threshold in Eq. (2) does

126 not automatically apply to every spatial or signed network.

127 **2.2 Physical identification and calibration**

128 Regular mesh refinement gives convergence of a discrete coupling-energy
129 quadrature to its continuum integral: scalar energy convergence, not conver-
130 gence of trajectories and not a construction of a physical kernel. The model
131 posits a continuum kernel and separately identifies its cortical realization
132 with endogenous electromagnetic coupling. The formal predicate requires
133 a jointly continuous kernel, a normalized substrate, a mean-field strength
134 equal to its spatial average, and no single site providing more than half of
135 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
136 model calibrated to cortex.

137 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
138 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
139 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
140 Fadden, 2020). These observations motivate an effective interaction without
141 establishing its magnitude relative to phase diffusion, its spatial averaging
142 rule, or its causal role in experience; changes in phase organization under
143 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
144 coupling threshold.

145 The physical-unit conversion is the sharpest open limitation of the frame-
146 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
147 shift and oscillation frequency, is dimensionless. It cannot be compared di-

rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \quad (4)$$

requires a rate calibration γ , where s is the time shift per unit field and E the field amplitude. Supplemental Material sets out why γ is not recoverable from the other factors: it states how a per-cycle phase advance accumulates into a drift rate, and the choice that would let it cancel removes the very ratio the threshold test is about. The saved field-strength calculations are conditional on this calibration and on the chosen spatial aggregation; they do not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel and phase diffusion from the same physical system is necessary before the threshold becomes a quantitative biological claim; Section 3.2 makes that measurement the first item of the proposed experiment.

3 Recovery after awakening: a conditional prediction

Sleep-associated changes in extracellular space and clearance motivate a possible slow control of coupling (Xie et al., 2013). That geometry is measurable and its cortical values are known: the extracellular space occupies a volume fraction $\alpha \approx 0.2$ of tissue at a diffusion tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008), and natural sleep raises α from roughly 0.14 to 0.23 (Xie et al.,

2013). It reaches Eq. (4) through the field amplitude E rather than through
a further mechanism: below 10 kHz cortical tissue conducts approximately
resistively at $0.3\text{--}0.6\text{ S m}^{-1}$, an order of magnitude set by the conductiv-
ity of saline diluted into that volume fraction (Barbour, 2017), so at fixed
transmembrane current density a larger α yields a smaller extracellular field.
We hypothesize that recovery after awakening includes a change in effective
coupling $K(t)$, mediated by extracellular geometry, on a slower timescale
than phase dynamics. The conductivity route fixes the sign that hypothesis
predicts—contraction of the extracellular space on awakening raises K —so
the design of Section 3.2 can fail it in direction as well as in magnitude. It
fixes nothing further: it is an effective-medium argument at fixed current
density, and is silent on how a heterogeneous field aggregates into the scalar
 K of Eq. (2). Neither the monotonic relation between geometry and cou-
pling nor its role in human sleep inertia follows from the cited physiology;
the model requires both to be measured.

If $K(t)$ rises through K_c and the phase distribution tracks the coherent
stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

The ideal stationary description has a subthreshold interval followed by con-
tinuous onset. Saturation requires a specified subsequent coupling trajectory;
monotonic coupling alone does not determine a logistic or any unique sig-

186 mold. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
 187 known. When coupling is inferred from the same coherence trace, this is not
 188 a parameter-free prediction of latency.

189 Critical slowing down limits the adiabatic prediction of Eq. (5) near
 190 threshold, and a finite population carries nonzero fluctuating order below
 191 it. An observable onset therefore requires a predeclared threshold above the
 192 finite-size floor, and a finite-rate trajectory need not show an infinite-slope
 193 cusp. Section 4.1 quantifies the limitation.

194 **3.1 Distributional and temporal tests**

195 The stationary model makes a second prediction: phase concentration and
 196 coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its appli-
 197 cability depends on the stationary-density assumption and the observation
 198 model, and the relation is common to the whole von Mises family, so agree-
 199 ment with it identifies neither ephaptic coupling nor this framework against
 200 alternative synchronization models. Within the same ansatz $a/r = K/D$ for
 201 $r > 0$, so the limiting threshold ratio follows from the density relation instead
 202 of testing it, and ratios near the finite-size floor are especially unstable.

203 The estimators must be independent, which is the constraint most easily
 204 violated in practice. Estimating a by inverting the Bessel ratio from r would
 205 enforce the relation being tested, and linear statistics on wrapped phases
 206 estimate neither quantity.

207 **3.2 A protocol for the awakening experiment**

208 The requirements above define a design rather than a single measurement,
209 and the design is where the content of the hypothesis lives. An adequate test
210 needs all six of the following.

- 211 **1. Coupling and diffusion in the same units.** Estimate K and D
212 from the same cortical regime, in inverse-time units, separating phase
213 diffusion from quenched frequency spread and from observation noise.
214 Without this the threshold $K > 2D$ is not an evaluable statement
215 about cortex.
- 216 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
217 an astrocytic signal is an indicator of the extracellular volume fraction
218 α , not a measurement of K . Its mapping to effective coupling must be
219 established separately, with its own uncertainty, and the conductivity
220 argument of Section 3 supplies only the expected sign of that mapping.
- 221 **3. Phase estimation at adequate spatial resolution.** At least 100
222 spatial samples per estimate. Estimate r by the circular resultant
223 length and a by an unweighted regression of log-density on the co-
224 sine of phase relative to the circular mean. With fewer sites, pooling
225 over time changes the estimand to a spatiotemporal distribution and
226 must be stated as such.
- 227 **4. An independent recovery endpoint.** Behavioral recovery measured

228 separately from the phase trace, so that latency is not defined by the
229 signal whose latency is at issue.

230 **5. A predeclared onset criterion.** A coherence threshold fixed above
231 the finite-size floor before unblinding, chosen in the knowledge that a
232 finite ramp smears the cusp.

233 **6. Model comparison under matched observation.** Coupling-gated
234 recovery compared against exponential, flexible sigmoidal and other
235 slow-recovery models under identical temporal filtering and observation
236 noise. Latency alone cannot distinguish the mechanism, because many
237 processes create delays.

238 The design fails informatively in three ways: failure of the stationary
239 phase-density relation over a sufficiently broad concentration range; absence
240 of the proposed coupling–geometry association; or a recovery trace better
241 explained by an alternative after finite-rate and observation effects are ac-
242 counted for. Two apparent failures are not failures. A linear apparent onset
243 is evidence against the adiabatic approximation only where that approxima-
244 tion is independently justified, and it does not falsify the field hypothesis
245 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
246 tion excludes stationary coexistence of two positive branches in this model,
247 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

248 4 Simulation results

249 The simulations characterize the finite-size, finite-rate regime that the sta-
250 tionary theory leaves open, and they bear on whether the prediction of Sec-
251 tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate
252 methodological issues that also arise in analyses of driven transitions and
253 dissipation-driven plasticity. Each is a finite run at its sampled parameters
254 and not a general statement, and none discharges a hypothesis of the com-
255 position in Section 5. Each run uses explicit seeds and compact saved sum-
256 maries, and the numerical values below are generated from those artifacts.
257 The supplement gives parameters, estimators, failure criteria and additional
258 panels. Identical-frequency runs use phase diffusion rather than a quenched
259 frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

260 4.1 Ramp speed sets the apparent onset exponent

261 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
262 four coupling speeds. Escape is the first of three consecutive saved samples at
263 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
264 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
265 with a fitted delay exponent 0.443. Their apparent onset exponents run from
266 0.898 to 0.417, against 0.443 to 0.448 returned by the same estimator on the
267 exact stationary branch sampled on the same coupling grids. At the fastest
268 speed the ensemble mean never reaches the declared fit window, so the value

the estimator returns there, 0.964, is a local log-slope of a delayed trajectory rather than an onset exponent.

The exponent recovered by fitting a driven transition is a joint function of the mechanism and the drive rate. An onset exponent read off a recovery trace therefore says little about the underlying transition unless the ramp rate is independently known and the finite-size floor is declared, and that holds for any such analysis, not only for one conducted under this framework. The prediction of Section 3 is the first thing it constrains: the stationary square-root law cannot be imposed on an arbitrary recovery trace, and its absence is not by itself evidence against a coupling-gated mechanism. The specified deviation-based critical ramp speed is not bracketed by the sweep.

The dynamical-selection control shows escape toward a coherent branch above threshold. At $K = 2.8$, steady mean order is 0.67735 against the static value 0.68270. Refining the integration step by a factor of 4 at fixed duration moves that order to 0.68168, so the bulk of the gap is Euler–Maruyama bias; the last refinement still moves it by 0.00147, so the remaining -0.00102 bounds the finite-size part rather than measuring it. The fitted growth-rate slope is 0.41505 against the theoretical $1/2$, over a window bounded below by the finite-size floor and above by the static branch; the same integrator bias enters that fit. The propagation-of-chaos diagnostics show decreasing subcritical dependence with increasing population size. In the supercritical unpinning ensemble, however, pair correlation remains substantial because realizations share a random collective orientation. This negative uncondi-

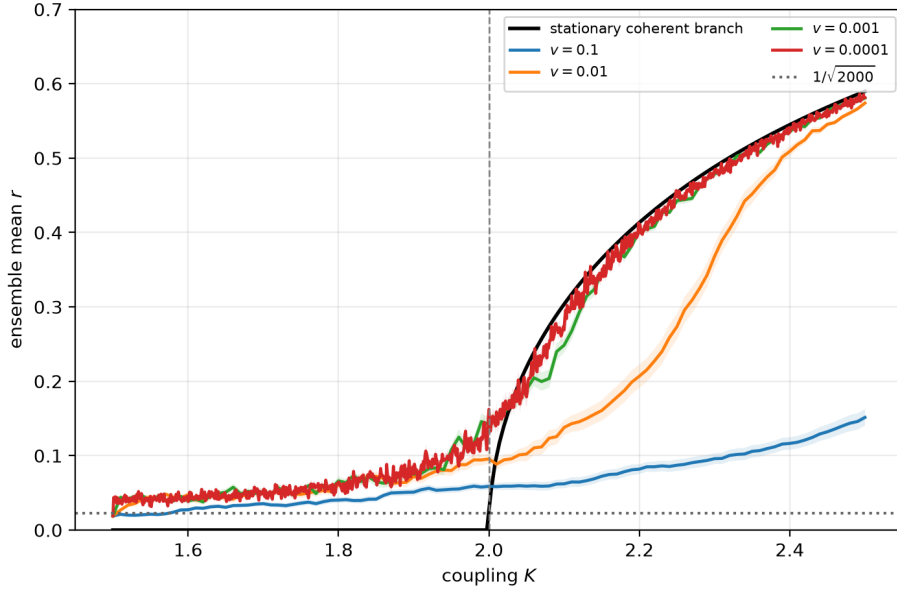


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

292 tional result does not settle a conditional mean-field limit after removal of
 293 that orientation.

294 4.2 Spatial coupling, frustration and calibration

295 A periodic two-dimensional sheet with row-normalized exponential coupling
 296 remains coherent in the sampled empirical decay-length band under the de-
 297 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
 298 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-

eter regime; fixing total coupling while changing range tests fragmentation,
not whether cortex supplies that total coupling.

A row-balanced Dale-law network shows that row balance alone does not
guarantee a floor-level incoherent baseline. In a separately specified weak-
synapse regime, a uniform modulatory term produces an operational coher-
ence crossing inside the bracket $K_{\text{eff}}/D \in [1.958, 2.009]$, from a coupling grid
refined to steps of 0.025 inside the one geometric step $[1.857, 2.110]$ that lo-
cates it. The refined bracket still contains the model’s own mean-field thresh-
old $K_c/D = 2$, so a tenfold finer grid does not separate the crossing from
the threshold at this size. Signed coupling amplifies finite-size fluctuations
in the same runs, whose incoherent baseline carries $N\langle r^2 \rangle > 1$, which is the
direction a resolved crossing would be displaced in; the refinement bounds
that displacement by the width of the bracket rather than exhibiting it. The
quantity K_{eff}/D is dimensionless; carrying it over to cortex needs the rate
calibration γ of Section 2.2, which neither this sweep nor the spatial control
supplies.

4.3 Squared-drift plasticity does not learn structure

The predictive dissipation bound of Section 5 bounds nonpredictive memory
by dissipation, and it is tempting to read a system that lowers its dissipation
as thereby coming to mirror the structure of its input. The bound licenses no
such reading: a memory carrying no information at all satisfies it with room
to spare, so lowering dissipation presses toward less representation rather

321 than toward better-aligned representation. This control exhibits a system in
 322 which the pressure is not merely toward less but away.

323 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

324 It is a quadratic functional of an affine image of the kernel. The formalization
 325 constructs the bounded drift operator A and its adjoint on the relevant L^2
 326 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 327 decreases this objective at fixed phases. The objective is not generally a
 328 thermodynamic entropy-production rate, which requires probability-current
 329 information. Whether biological couplings follow this flow is an additional
 330 modelling question.

331 The joint control evolves noisy phases with clustered frequencies and
 332 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 333 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 334 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so
 335 both endpoints contribute. Resource constraints and evolving phases re-
 336 move the fixed-phase monotonicity guarantee. Each seed runs three arms on
 337 one phase-noise stream: the gradient step, a symmetric random step of the
 338 same Frobenius norm, and a frozen kernel. Denying an implication requires
 339 exhibiting its antecedent, so the reported quantity is how much of the ob-
 340 jective the update actually removes: symmetric coupling cancels from the

341 mean drift, which fixes a floor of 990 beneath the frozen arm's 1380.6063 to
 342 1382.3448, and the gradient arm removes between 0.6773 and 0.6914 of that
 343 headroom. The random arm of matched step size removes none of it, so the
 344 descent belongs to the gradient. The random arm, however, ends an order
 345 of magnitude less deformed than the gradient arm: successive isotropic steps
 346 cancel, leaving cumulative norm growth of 1.2879 to 1.3056 against the gra-
 347 dient arm's 12.4094 to 14.6958. Because the random arm barely enters the
 348 sparsification channel, it does not separate the gradient descent from cumu-
 349 lative deformation. The learning rate 0.2 is the rate at which the objective
 350 descends furthest across a sweep from 0.001 to 0.4 evaluated on one of the
 351 three reported seeds, not a rate at which the update is known to converge.

352 Across three seeds, second-half order lies between 0.9167 and 0.9222: co-
 353 herence survives the descent. Alignment with the cluster template is read
 354 from the ratio of mean within-cluster to mean between-cluster coupling,
 355 which is unchanged by rescaling the kernel. That ratio starts between 0.9824
 356 and 0.9948 and ends between 0.3507 and 0.6135, while the random arm leaves
 357 it at 0.9965 to 0.9997; against 2000 node relabellings the final kernel is fur-
 358 ther from the true template than from a fraction 0.9510 to 1.0000 of them
 359 (Fig. 3). The reference for that fraction is the frozen arm on the same seeds,
 360 which reaches 0.6435 to 0.9110 without any update at all, and not a null
 361 median of one half; the increment the gradient step contributes is the differ-
 362 ence between those ranges, which at three seeds is small enough that they
 363 nearly meet. Two readouts do not carry the effect and we do not rest it

364 on them. The initial-minus-final Frobenius distance to the template lies be-
 365 tween -58.0962 and -48.1760 , so that distance grows; but the kernel norm
 366 grows by a factor of 12.4094 to 14.6958 over the same interval, and a fixed
 367 total resource constrains the sum of the entries rather than the norm, so
 368 most of that growth is concentration of mass onto fewer edges rather than
 369 loss of alignment. The edge-level correlation with the template ends between
 370 -0.0362 and -0.0208 , near its starting value: at this sparsity the edge-level
 371 variance swamps a contrast that the block means resolve. The template is a
 372 covariance of shared latent cluster signals, not an input to the update or a
 373 measured covariance of the constant frequencies.

374 The objective descends by most of what is available to it, coherence is
 375 maintained, and within-cluster coupling falls to a fraction of between-cluster
 376 coupling. Coherence is therefore not a proxy for representational alignment
 377 in this regime, and a plateau in the objective establishes neither stationar-
 378 ity of the adaptive process nor a thermodynamic entropy-production rate.
 379 Nor is the direction of the failure accidental. Symmetric coupling conserves
 380 the mean drift, so minimizing $\sum_i v_i^2$ can only equalize drift across nodes,
 381 and equalizing drift rewards coupling between clusters of unlike frequency—
 382 the complement of a template that rewards coupling within clusters of like
 383 frequency. The objective and the template are in this pairing opposed by con-
 384 struction, and that is the point: nothing about a system’s dissipation fixes
 385 which structure its slow variables acquire. What fails here is the objective
 386 rather than thermodynamically specified learning at large: an Ising model

387 whose couplings are annealed degrees of freedom, driven by external fields
388 that carry the data, memorizes and generalizes under ordinary statistical-
389 mechanical rules (Corberi et al., 2026). Those couplings equilibrate under
390 a Hamiltonian containing the data; these descend a squared-drift functional
391 whose stationary point the data never enters. Which functional the slow
392 variables follow is what decides whether they acquire environmental struc-
393 ture.

394 **4.4 The estimation protocol demonstrated on public** 395 **EEG**

396 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
397 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
398 et al., 2025). The purpose is to show what the protocol produces and where
399 it is fragile, not to test the awakening hypothesis, for which these data are
400 the wrong design.

401 Two fragilities appear immediately. The implemented estimator pools 62
402 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
403 tion rather than an instantaneous spatial one—a departure from the protocol
404 forced by channel count, and one that any scalp implementation will face.
405 Bipolar and common-reference montages give different residuals, making ref-
406 erencing a material analysis choice rather than a preprocessing detail.

407 The bipolar results are compatible with the Bessel relation over the ob-

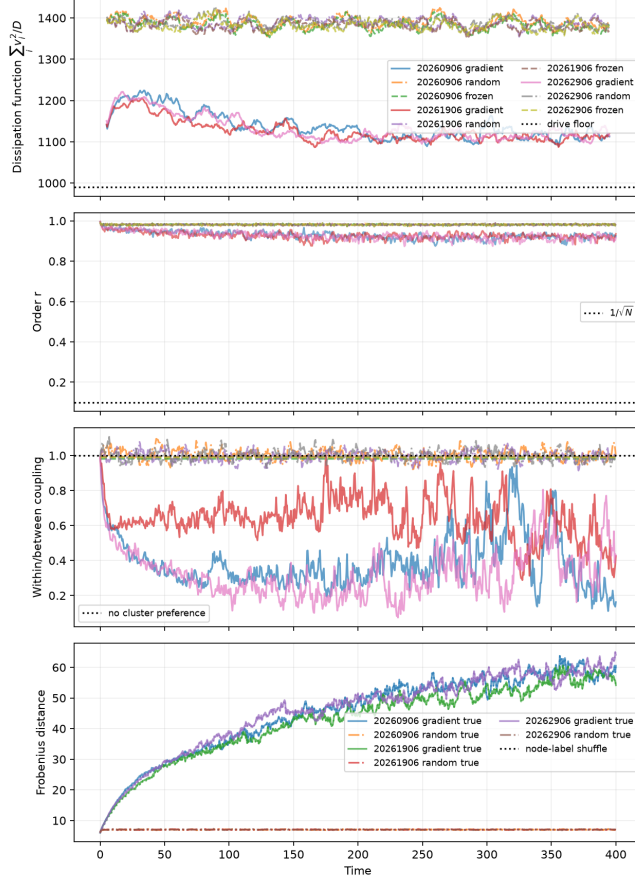


Figure 3: Joint fast/slow dynamics, three arms per seed on one phase-noise stream: gradient (solid), matched-norm random step (dash-dotted), frozen kernel (dashed). First: smoothed squared-drift dissipation against the drive floor the conserved mean drift imposes; only the gradient arm descends. Second: order and the finite-size reference. Third: mean within-cluster over mean between-cluster coupling, a measure unchanged by rescaling the kernel; the gradient arm falls well below the no-preference line while the other arms stay on it. Fourth: Frobenius distance to the true template (solid) and to a node-label shuffle (dotted), which track each other because the kernel norm, not the alignment, is what grows. Coherence persists while alignment is lost. The result limits the proposed connection between dissipation minimization and environmental structure.

served concentration range. That range is too narrow to distinguish the curve from its linear approximation at the observed scatter, so the check is uninformative about the relation itself. An informative test needs a broader concentration range and explicit treatment of source mixing, finite samples and temporal pooling; intracranial or high-density recordings across a genuine state transition are the natural setting.

5 Thermodynamic constraints and the formal composition

The synchronization model of Section 2 is one link of a longer conditional construction, whose nine nodes are:

n1 a finite phase space has bounded information capacity; **n2** a broken symmetry forces a boundary, where the field leaves its vacuum manifold; **n3** the boundary erases, and erasure costs heat; **n4** that dissipation bounds nonpredictive memory; **n5** discrete couplings coarse-grain to a continuum limit; **n6** a specified physical field realizes that limiting coupling; **n7** above threshold the field carries a unique coherent order parameter; **n8** compatible local states glue to a unique global section; **n9** that section is the unique fixed point of a self-prediction map.

Node-level results are established or cited; what the composition consumes are the eight implications between them. Figure 4 shows where they sit and what kind of assumption each one is, and Table 1 states them, classified by their role in the composition rather than by their plausibility. In particular,

431 finite capacity does not imply a symmetry-breaking boundary: n_1 and n_2 are
 432 independent roots, even though the composition can accept an implication
 433 between them as an argument.

434 5.1 Capacity, boundaries and predictive memory

435 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
 436 equality only for the uniform distribution. A stream with more possible
 437 histories than $|X|$ cannot be encoded injectively in that register. Separately,
 438 a continuous field on a connected substrate that reaches distinct components
 439 of a disconnected vacuum manifold must leave that manifold somewhere,
 440 the simplest topological-defect construction (Kibble, 1976). Relating this
 441 locus to a particular finite information-processing register is the additional
 442 hypothesis E12 carries.

443 A non-surjective update on a finite register is non-injective. Under the
 444 specified system–bath model and thermal assumptions, erasure carries a Lan-
 445 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 446 update erases, and logical many-to-one processing must be distinguished from
 447 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 448 tion records the physical heat identity as an obligation of each model and
 449 supplies an explicit one-bit eraser satisfying it.

450 For a system state X_t driven by an environmental signal S_t , the predictive

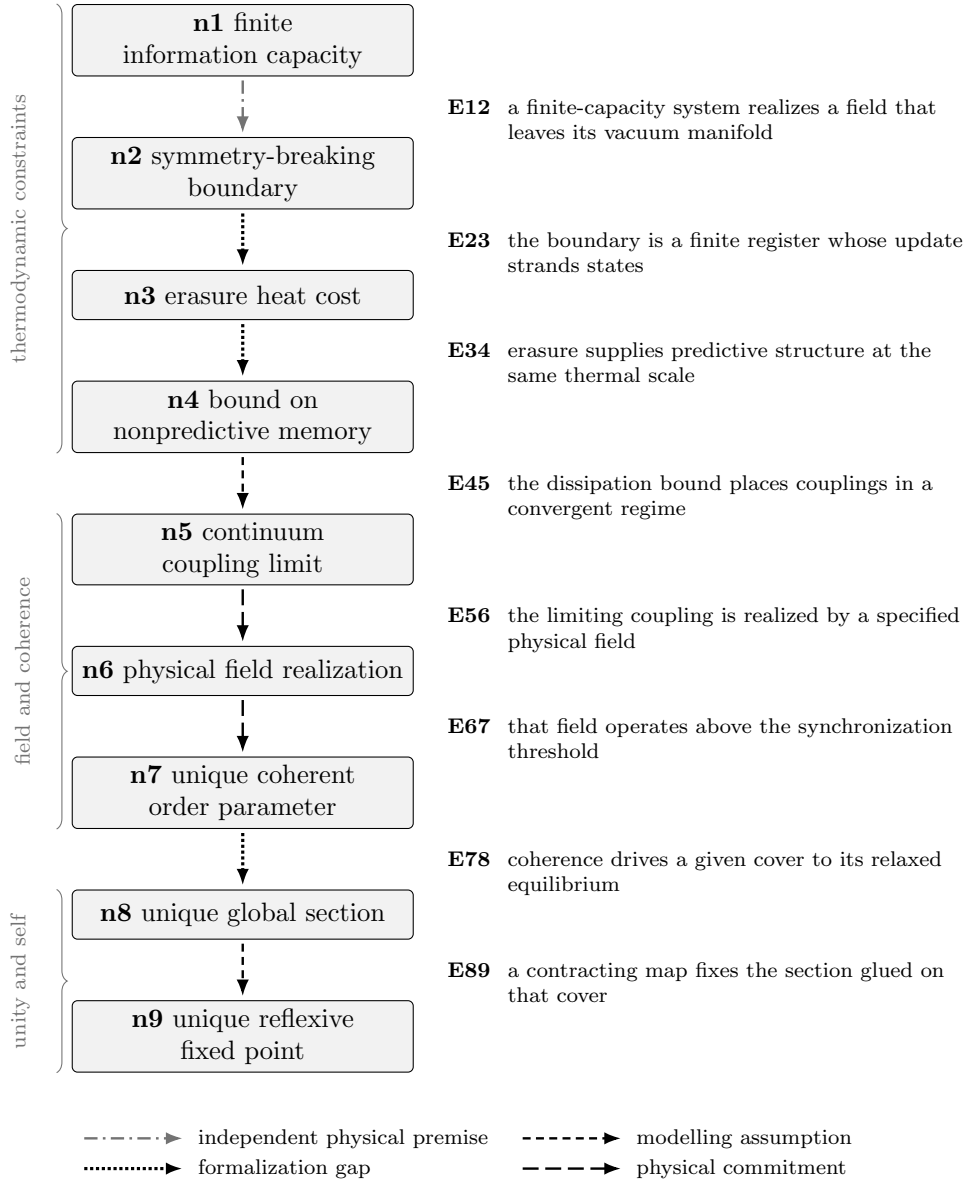


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12	Independent	A finite-capacity system realizes a field that
capacity $\rightarrow$	physical	leaves its vacuum manifold. Capacity alone
boundary	premise	does not imply this.
E23	Formalization	The boundary is associated with a finite
boundary $\rightarrow$	gap	register whose update leaves states
erasure		unreachable.
E34	Formalization	Erasure supplies a predictive structure with
erasure $\rightarrow$	gap	the same thermal scale and heat budget,
prediction		and nonzero wasted memory.
E45	Modelling	A predictive dissipation bound places
prediction $\rightarrow$	assumption	coupling energies in a convergent
continuum		coarse-graining regime.
E56	Physical	The limiting coupling is realized by a
continuum $\rightarrow$	commitment	specified field satisfying the kernel
field		conditions. Its identification with cortex is
		empirical.
E67	Physical	That field operates above the
field $\rightarrow$	commitment	synchronization threshold.
coherence		
E78	Formalization	Coherence drives a cover, supplied as a
coherence $\rightarrow$	gap	separate argument of the theorem, to an
unity		equilibrium reached by relaxation at
		coupling at least the named mean-field
		value. That its patches carry compatible
		local states is a field of the cover rather
		than a conjunct of this edge.
E89	Modelling	The state glued on that same cover is fixed
unity $\rightarrow$ self	assumption	by a contracting map mediated by its local
		self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

451 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

452 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
453 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
454 nonpredictive memory at a given dissipation budget. It does not specify
455 a learning rule, maximize prediction, or require the internal representation
456 to reproduce environmental geometry—the point the plasticity control of
457 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
458 ditionally requires nonpredictive information to be bounded by the entropy
459 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
460 bound under that identification; it does not derive the identification.

461 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
462 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
463 tropy, not a general identification of relative entropy with physical entropy
464 production. The chain uses a different technical sense of information at each
465 step—a counting bound on a register, a thermodynamic heat cost, a predic-
466 tive mutual information—and those senses are not interchangeable (Rosas
467 et al., 2026). These distinctions prevent an informational inequality from
468 being read as a derivation of environmental learning, Bayesian inference, or
469 semantic content.

470 5.2 Compatible states and reflexive fixed points

471 A sheaf formalizes local descriptions and their restriction to smaller regions.
472 If local sections agree on every overlap, the sheaf condition gives a unique
473 global section. In the probability-sheaf construction, this is consistency of
474 local density information, and it is not implied by agreement of oscillator
475 phases: the development includes a counterexample with equal phases but
476 incompatible densities on an overlap. Section 6 rests on this distinction,
477 and the model excludes such configurations only through its compatibility
478 hypothesis.

479 A thermodynamic cover couples this gluing construction to a phase con-
480 figuration at a potential minimum. For the deterministic positive-coupling
481 model, convergence to that minimum is proved under an explicit initial basin
482 condition; other equilibria prevent an unconditional conclusion. The station-
483 ary noisy order parameter, the deterministic relaxation, and the cover must
484 therefore be connected by E78 rather than identified without a hypothesis.
485 Neither field continuity nor synchronization alone guarantees a globally com-
486 patible representation.

487 A reflexive extension equips global sections with a complete metric and a
488 predictive map formed by restricting a state to a local encoding region and
489 reading that encoding out. If the map contracts and fixes the section supplied
490 by the cover, Banach's theorem gives uniqueness of that same fixed point.
491 The contraction rate is assumed, and the equality between the glued section
492 and the fixed point is part of E89. A coherent order parameter supplies the

493 supercritical inequality required by the stated contraction bound; it does not
494 construct the map or prove that cortical dynamics implements it.

495 The theorem `chain` concludes `UnifiedSelf` from these hypotheses and
496 from two structures it also takes as arguments, a thermodynamic cover and
497 a reflexive boundary. The eight edges are not the whole of what is assumed:
498 the cover’s overlap-compatibility field—the load-bearing assumption of the
499 unity-is-not-synchrony argument—and the boundary’s resonance condition
500 are class fields, so they are consumed by the theorem without appearing
501 among its edges. A single witness satisfies all eight at once, so their conjunc-
502 tion is not empty; it is assembled from a double well, a register, a uniform-grid
503 mesh and a three-site cortex, which remain distinct toy systems rather than
504 one mechanism. The project declares no additional Lean axioms beyond
505 the foundational axioms used by Lean’s libraries; physical content resides
506 in hypotheses and model fields. This checks the conditional mathematical
507 composition, not the empirical truth of its premises. Detailed constructions,
508 scope statements and counterexamples are retained in the supplement and
509 source files.

510 **6 What the construction commits to about** 511 **consciousness**

512 The composition terminates in a mathematical object, and we take a posi-
513 tion on what that object is. The commitment has two clauses. *Unity*: the

514 unique global section glued from compatible local states is the formal corre-
515 late of the unity of a conscious episode—the fact that a subject undergoes
516 one experience rather than a collection of co-occurring local ones. *Selfhood*:
517 the fixed point of the self-prediction map on that same section is the formal
518 correlate of a minimal self, a state that is what it takes itself to be. Neither
519 clause is a theorem, and no simulation here supports either. We argue for
520 them, then say what would count against them.

521 Three features of the construction motivate the identification, and each
522 distinguishes it from a bare synchrony measure.

523 First, compatibility rather than agreement. Oscillator phases can agree
524 everywhere while the local probabilistic descriptions they carry remain in-
525 compatible on their overlaps, and Section 5.2 exhibits that configuration
526 inside the formalization, not only as an illustration. If the unity of experi-
527 ence is a matter of local contents being about one world rather than merely
528 co-timed, then compatibility, not coherence, is the property to look for—and
529 coherence enters as the physical condition under which compatibility can be
530 maintained, which is what E78 asserts.

531 Second, uniqueness follows from the construction rather than from a sep-
532 arate selection principle. Accounts of conscious unity need something that
533 picks one experience from among the candidates a system could support;
534 integrated information theory postulates exclusion for this purpose (Tononi
535 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
536 whose local sections agree, the sheaf condition gives exactly one global sec-

tion, and given a contracting self-prediction map, Banach’s theorem gives exactly one fixed point. The scope of that claim needs stating exactly, because the stronger reading is available and is not ours. Uniqueness is relative to a cover and a map, and choosing them—which regions, which overlaps, which encoding region, and so where the subject ends—is not supplied by either theorem. It is what E78 and E89 carry. What the construction removes is the need for an additional principle to select among the experiences a fixed cover and map would otherwise admit; it does not remove the need to say which cover and map a given subject has. That is a structural argument, not evidence.

Third, the reflexive clause makes the self a condition rather than an addition. The predictive map restricts a global state to a region encoding it and reads that encoding back out, so a fixed point is a state whose self-model is accurate about the state it models. That condition is available only because the state modelled is a single glued object: a system whose local descriptions do not cohere has no such state for a self-model to be about. Unity and selfhood are therefore one construction read at two levels, which is why `UnifiedSelf` requires the fixed point to be the glued section itself rather than an unrelated section.

The costs belong with the argument. The commitment does not address why any physical condition is accompanied by experience; it is a proposal about which physical state is the conscious one, not about why states of that kind feel like anything. It is therefore open to the objection that a system

560 could satisfy every condition—compatible local states, a supercritical field,
561 an accurate self-model—while nothing is experienced. We have no reply to
562 that objection and claim none. The identification is also multiply realizable
563 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
564 or to electromagnetic coupling, so the substrate claim is carried entirely by
565 E56 and E67 and is empirical. No conclusion about artificial systems follows
566 from the formal results.

567 What could count against the commitment is dissociation. It predicts that
568 the two clauses and the presence of a unified conscious episode go together,
569 so a state in which local cortical descriptions are demonstrably compatible
570 and the field supercritical, while the subject reports no unified experience,
571 would tell against it; so would unified reported experience where compati-
572 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
573 states are the places to look, because they separate coherence from report
574 in ways ordinary waking does not. Such tests need an operational measure
575 of overlap compatibility on neural data, which this article does not supply;
576 constructing one is the main obstacle to making the commitment empirically
577 live rather than merely stated. Supplemental Material specifies what such an
578 observable would have to satisfy—a restriction map, a statistic with a null
579 under decoding noise, validation against configurations whose compatibility
580 is built rather than inferred—and the three ways it would fail silently, one
581 of which is reading compatibility off synchrony and so assuming away the
582 distinction being tested.

583 The physical tests of Sections 3 and 4 are independent of all of this. They
584 concern coupling, coherence and recovery, and can fail while the identification
585 stands, or succeed while it is wrong.

586 **7 Discussion**

587 The framework connects thermodynamic information constraints, collective
588 phase organization and compatible local representations, conditionally and
589 with its assumptions named. Its biological content is a coupling-gated re-
590 covery hypothesis with defined observables, a stated protocol, and failure
591 conditions declared in advance. Its formal content is a composition whose
592 eight gaps are exhibited rather than argued past, and which can be inhabited
593 by a single model. Its interpretive content is the identification of Section 6,
594 offered as a commitment with reasons.

595 The simulations are useful because they restrict the hypothesis rather
596 than support it. Finite ramps smear the stationary onset to the point where
597 a fitted onset exponent chiefly reports the ramp; an unpinned collective
598 phase makes independence diagnostics ambiguous in exactly the supercritical
599 regime the model cares about; and squared-drift plasticity maintains coher-
600 ence while degrading structural alignment. The second and third of these
601 are results about method and about a common intuition, and they transfer
602 beyond this framework.

603 The proposal differs in emphasis from accounts based on integrated in-

604 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
605 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
606 candidate interaction medium and a mathematical compatibility condition,
607 and, relative to a given cover and self-prediction map, it derives the unique-
608 ness those accounts postulate. It does not derive the exclusion principle of
609 integrated information, identify every broadcast with a field transition, or
610 derive a generative model from a dissipation bound. Continuous fields can
611 carry incompatible local descriptions and discrete models can admit sheaf
612 constructions, so continuity alone does not settle the combination problem;
613 compatibility is what this account offers in its place.

614 Two formal results bear on the substrate question, and both admit a
615 stronger reading than they support. At matched coupling resource, a fixed
616 wiring support that excludes a better-correlated pair can be suboptimal—a
617 statement about constrained optimization, not about biology. On an atom-
618 less substrate, a kernel supported on finitely many points contributes zero to
619 the chosen continuum coupling integral. That is a property of the measure
620 and the observable, and it cuts against this framework’s own coarse-graining
621 step as much as against anything else. It does not imply that digital hard-
622 ware cannot approximate a field or support consciousness, and no biological-
623 versus-artificial conclusion follows.

624 The empirical priorities follow directly from Section 3.2: estimate effec-
625 tive coupling and diffusion in consistent units, test the proposed relation
626 between coupling and extracellular geometry, and measure recovery against

an independently defined behavioral endpoint under matched model comparison. A favorable concentration-coherence relation would support a statistical description, not identify its physical cause; an unfavorable result in a regime satisfying the stated assumptions would constrain that description. Successful synchronization simulations cannot substitute for measurements distinguishing ephaptic interactions from other coupling mechanisms.

The article therefore asks to be judged in three separable parts: a prediction that can be tested, a composition that can be checked, and an identification that can be argued about. Keeping them separate is what makes the whole assessable.

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the manuscript source repository. Publication simulation values are regenerated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); preprocessing and selection are described in Supplemental Material. The source repository contains the reproducible analysis and does not redistribute the raw recordings.

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646 This research received no external funding. The author declares no compet-
647 ing interests.

648 **Use of AI tools**

649 AI assistance to this work was substantial and is not separable by task.
650 Several coding assistants and chat interfaces were used in an interleaved way
651 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
652 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
653 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
654 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
655 they contributed to the Lean development, the simulation and analysis code,
656 and the drafting and revision of both documents. Because the tools were
657 used in combination and in alternation, no section, proof, figure or result is
658 attributable to any one of them, and none is claimed to be. The author is
659 responsible for all content, including everything a tool produced. Machine
660 checking establishes the stated Lean results under their hypotheses; it does
661 not validate biological interpretations or replace scientific review of generated
662 code and prose.

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